Project References

This file collects research papers and foundational references that are directly relevant to the project. The list is grouped by the part of the system it supports.

1. Multimodal Fusion and Transformers

1. Attention Is All You Need (2017)
   - Why it matters: the project uses transformer encoders for text and vision variants.
   - Link: https://arxiv.org/abs/1706.03762

1. MSAF: Multimodal Split Attention Fusion (2020)
   - Why it matters: relevant to the project's attention-based multimodal fusion design.
   - Link: https://arxiv.org/abs/2012.07175

1. Attention Bottlenecks for Multimodal Fusion (2021)
   - Why it matters: useful background for compact multimodal fusion strategies.
   - Link: https://arxiv.org/abs/2107.00135

1. Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization (2016)
   - Why it matters: the project uses Grad-CAM-style image explanations for handwriting models.
   - Link: https://arxiv.org/abs/1610.02391

2. Speech and Audio Representation Learning

1. wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations (2020)
   - Why it matters: a strong reference for the project's self-supervised audio pretraining path.
   - Link: https://arxiv.org/abs/2006.11477

1. Interpretable Deep Learning Model for the Detection and Reconstruction of Dysarthric Speech (2019)
   - Why it matters: useful background for the speech feature branch and interpretable speech analysis.
   - Link: https://arxiv.org/abs/1907.04743

3. Reading, Dyslexia, and Eye-Tracking

1. RADAR: A Novel Fast-Screening Method for Reading Difficulties with Special Focus on Dyslexia (2016)
   - Why it matters: a strong reference for eye-tracking-based reading difficulty screening.
   - Link: https://arxiv.org/abs/1611.05669

1. Reading ability detection using eye-tracking data with LSTM-based few-shot learning (2024)
   - Why it matters: relevant to the project's visual focus and eye-tracking analysis path.

- Link: https://arxiv.org/abs/2409.08798

1. An Approach to Track Reading Progression Using Eye-Gaze Fixation Points (2019)
   - Why it matters: useful for understanding fixation-based reading progression and gaze metrics.
   - Link: https://arxiv.org/abs/1902.03322

1. Let AI Read First: Enhancing Reading Abilities for Individuals with Dyslexia through Artificial Intelligence (2025)
   - Why it matters: directly relevant to assistive reading support for learners with dyslexia.
   - Link: https://arxiv.org/abs/2504.00941

4. Handwriting and Dysgraphia

1. Automated dysgraphia detection by deep learning with SensoGrip (2022)
   - Why it matters: relevant to handwriting-based learning-disorder screening and early intervention.
   - Link: https://arxiv.org/abs/2210.07659

1. Handwriting Anomalies and Learning Disabilities through Recurrent Neural Networks and Geometric Pattern Analysis (2024)
   - Why it matters: useful background for handwritten anomaly analysis and learning-disorder inference.
   - Link: https://arxiv.org/abs/2405.07238

1. Towards Accessible Learning: Deep Learning-Based Potential Dysgraphia Detection and OCR for Potentially Dysgraphic Handwriting (2024)
   - Why it matters: supports the project's handwriting recognition and dysgraphia-related research direction.
   - Link: https://arxiv.org/abs/2411.13595

5. Related Supportive References

1. Multimodal Fusion Learning with Dual Attention for Medical Imaging (2024)
   - Why it matters: not a dyslexia paper, but useful for understanding dual-attention fusion patterns.
   - Link: https://arxiv.org/abs/2412.01248

1. Attention-guided Multi-step Fusion: A Hierarchical Fusion Network for Multimodal Recommendation (2023)
   - Why it matters: provides another useful attention-fusion reference for multimodal design.
   - Link: https://arxiv.org/abs/2304.11979

6. How To Use These References

- Use the multimodal fusion and transformer papers when discussing the core architecture.

- Use the eye-tracking papers when discussing the visual focus test and gaze-based metrics.
- Use the handwriting and dysgraphia papers when discussing handwriting analysis and biomarkers.
- Use the speech papers when discussing audio pretraining and spoken-response support.
- Use the assistive reading paper when discussing intervention or user-facing reading support.

7. Related Documentation

For the current project-specific evaluation flow, also see:

- docs/MODELCATALOG.md (/d:/Project/DyslexiaDetectionSystem/docs/MODELCATALOG.md)
- docs/furtherresearch/experimentmatrix.md (/d:/Project/DyslexiaDetectionSystem/docs/furtherresearch/experimentmatrix.md)
- docs/PROJECTDOCUMENTATION.md (/d:/Project/DyslexiaDetectionSystem/docs/PROJECTDOCUMENTATION.md)

8. Notes

- Some papers are direct project matches.
- Some papers are adjacent references that support the same architectural ideas.
- If you prepare a paper or thesis, it is best to cite the most directly relevant papers from each section rather than all of them.